Considerations for Effective Exploratory Data Analysis

Welcome to the lesson on the considerations for effective exploratory data analysis. Exploratory data analysis, or EDA, is a crucial step in the data analysis process.

Data Understanding

01 Ensure a clear understanding of the data's context to align with intended objectives

02 Thoroughly comprehend the dataset, its origins, and its relevance to the analysis goals

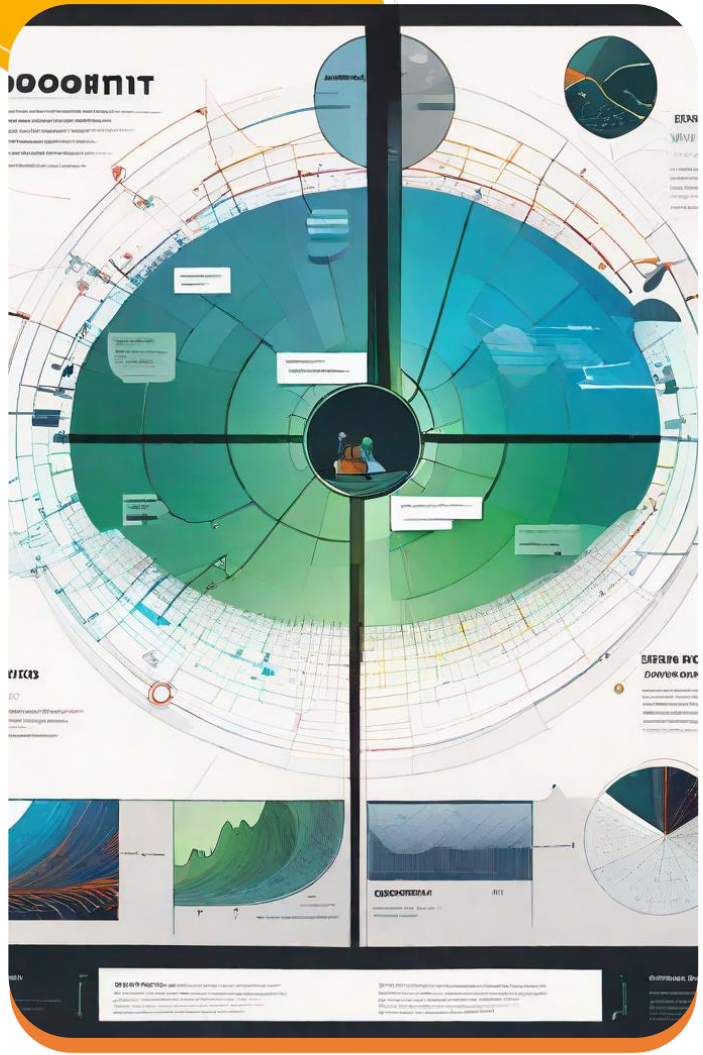
03 Examine data sources, understand variables, and identify any potential biases or limitations

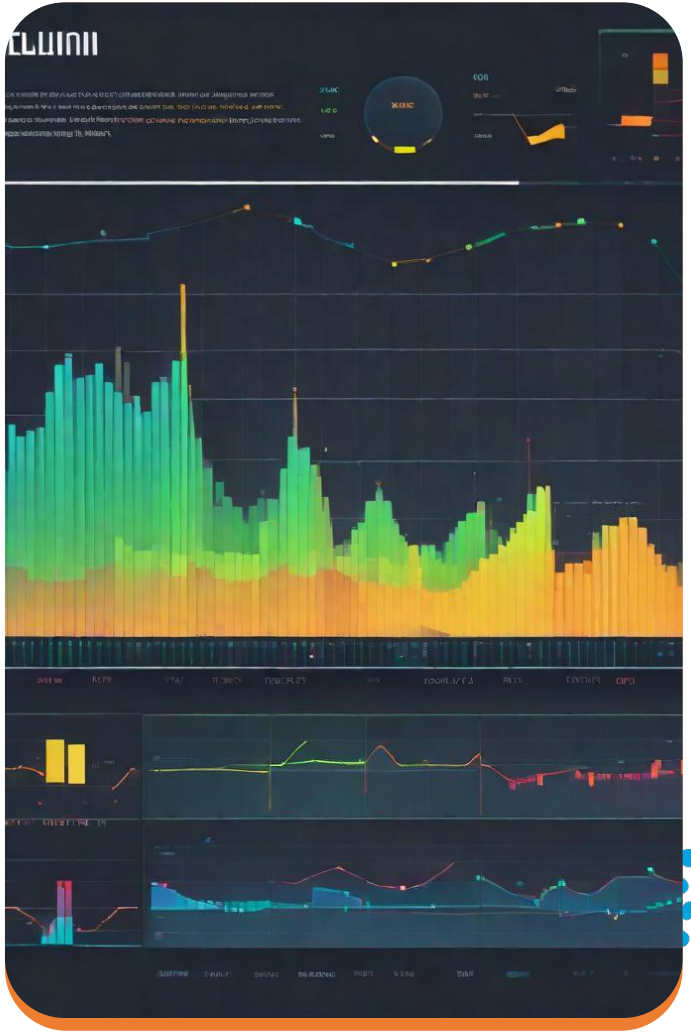
Iterative Process

01 Uncover deeper insights and refine understanding over time

02 EDA is an iterative cycle of exploration, analysis, and refinement

03 Insights prompt new questions and hypotheses





Visualization Choices

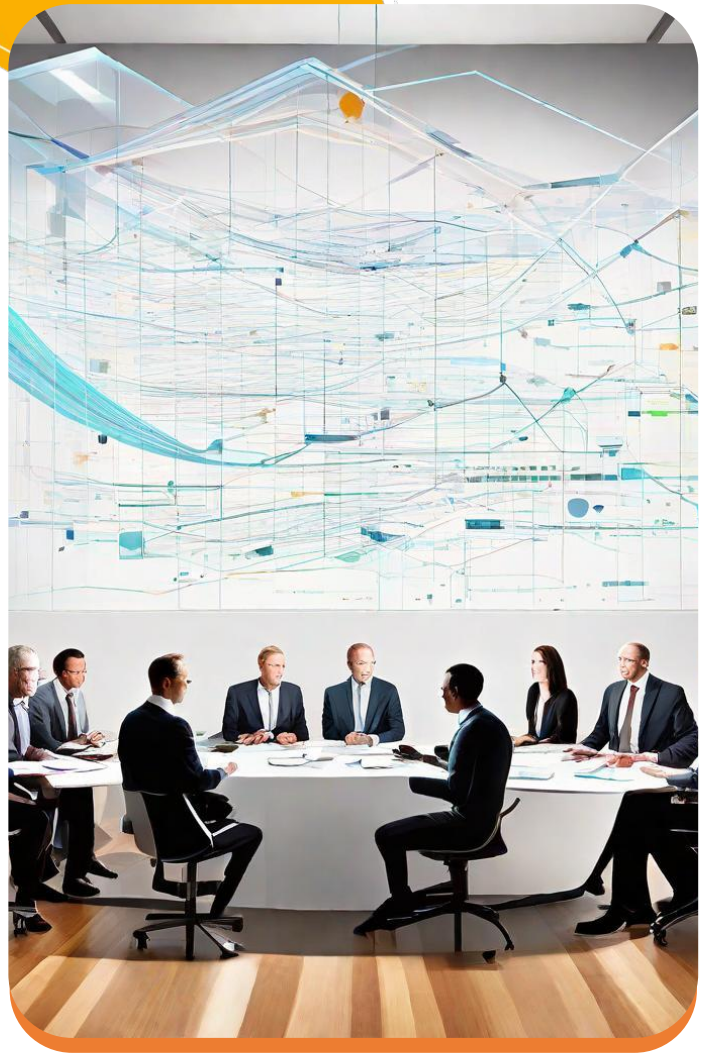
- Choose appropriate visuals based on data's nature
- Histograms, scatter plots, heatmaps, or interactive visualizations
- Enhance understanding and identify trends, outliers, and relationships

Domain Knowledge

01 Leverage industry-specific trends, nuances in data collection, and complex relationship interpretation

02 Incorporate domain expertise to enrich interpretation of findings

03 Contextualize insights, validate assumptions, and form meaningful hypotheses

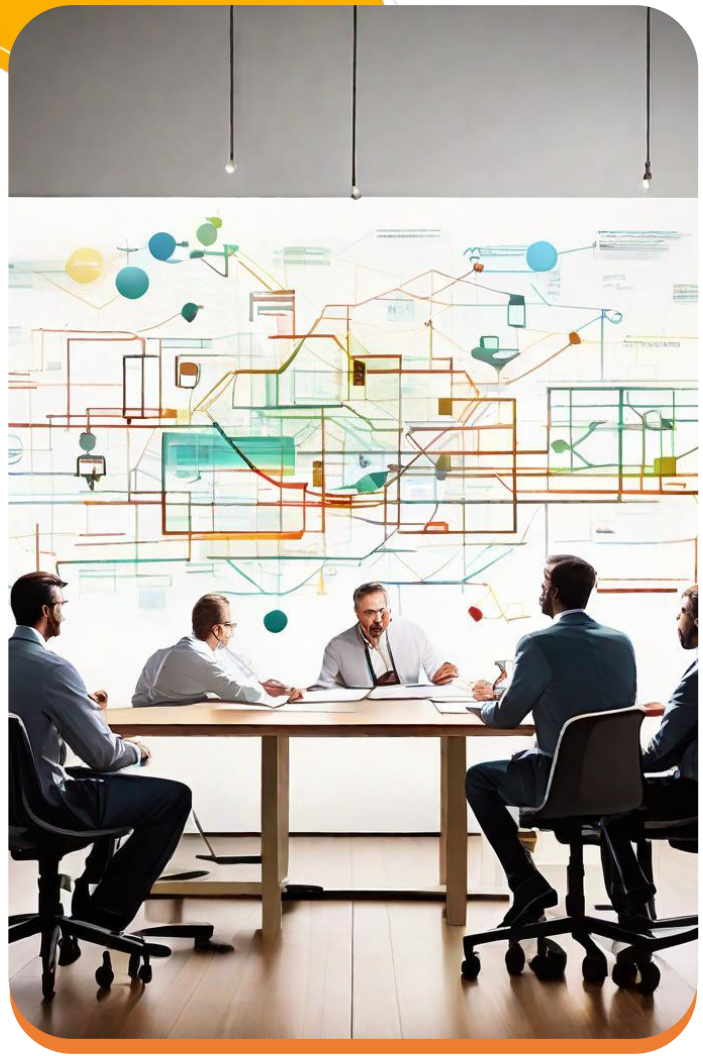


Conclusion

01 Stay aligned with necessary insights.

02 Leveraging domain knowledge strengthens the analytical process

03 Foster more insightful conclusions



Thank you. 😄